# Lab: Clean the OrbitCo Incident Export

**Type:** Hands-on lab (individual work) **Time:** 20 minutes (15 min individual work + 5 min debrief)

## Objective

Take a raw, messy 420-row incident export and produce a clean, analyzable `tickets_clean` Excel Table that preserves the raw data as evidence and documents every cleanup decision.

## Scenario

You have received `orbitco_tickets_raw.csv` — a 420-row export from the OrbitCo ticketing system covering the last 30 days. Your director wants a summary by tomorrow morning. Before you can compute anything, the file needs work. A quick visual scan turns up:

- A stray merged-header row pasted in at row 47 (someone else's copy-paste mistake)
- Seven duplicate ticket IDs caused by a reopen-bug in the ticketing system
- Inconsistent date formats across the `OpenedAt` and `ClosedAt` columns
- Nineteen rows with a blank `Severity` (the support agent resolved and closed them before triage)
- Trailing whitespace on six ticket IDs
- Four mixed-case severity values ("high", "High ", "HIGH") that will break `COUNTIF` later

## Instructions

1. Open a blank Excel workbook. Use **Data → From Text/CSV** to import `orbitco_tickets_raw.csv` onto a sheet named `tickets_raw`. Do not edit this sheet at any point in this lab. It is your evidence.
2. Right-click the `tickets_raw` tab, choose **Move or Copy**, check "Create a copy," and rename the new sheet `tickets_clean`. All cleanup happens on this sheet.
3. Scroll to row 47 on `tickets_clean` and delete the stray merged-header row (Home → Delete → Delete Sheet Rows).
4. Select the `OpenedAt` column, then **Data → Text to Columns → Delimited → Next → Next → Column Data Format: Date (MDY) → Finish**. Repeat for the `ClosedAt` column. This actually converts the underlying type — simply reformatting the cell does not.
5. In an empty column, add a helper formula `=TRIM([@TicketID])` to strip trailing whitespace from ticket IDs. Copy the helper column, paste it back onto the `TicketID` column as **Values only** (Paste Special → Values), then delete the helper column.

6. In an empty column, add a helper formula `=PROPER(TRIM([@Severity]))` to normalize case and whitespace on severity (so "high", "High ", and "HIGH" all become "High"). Paste-as-values back onto the `Severity` column, then delete the helper.
7. Select any cell in the data range and press **Ctrl+T** (or **Cmd+T**). Confirm "My table has headers" and click OK. On the **Table Design** tab, rename `Table1` to `Tickets`.
8. Select the `Severity` column in the Tickets Table. **Data → Data Validation → Settings → Allow: List → Source: `High, Medium, Low`**. This prevents future typos at entry time.
9. Add a new column to the right of the Table named `SeverityIsMissing` with the formula `=ISBLANK([@Severity])`. Do not delete the rows where Severity is blank. Flagging preserves the accurate total ticket count.
10. With any cell in the Table selected, **Data → Remove Duplicates → uncheck all columns → check only `TicketID` → OK**. Excel reports the number of duplicates removed. Confirm it says **7 duplicate values found and removed**. If the number is not exactly 7, the dedupe key is wrong — undo and check that you selected only the `TicketID` column.
11. Add a computed column `ResolutionMinutes` with the formula `=([@ClosedAt] - [@OpenedAt]) * 24 * 60`. This gives you resolution time in minutes (the correct unit for operations reporting).
12. Save the file as `orbitco-workbook-[yourname].xlsx`. You will extend this same file in the next lab.

## Example

Before cleanup, a sample row in `tickets_raw` might look like:

| TicketID | OpenedAt | ClosedAt | Severity | Team |
|---|---|---|---|---|
| `T-10047` (trailing space) | `"4/2/2026 9:12"` (text) | `"4/2/2026 11:03"` (text) | `"HIGH"` | Platform |

After cleanup, the same row in the `Tickets` Table reads:

| TicketID | OpenedAt | ClosedAt | Severity | Team | SeverityIsMissing | ResolutionMinutes |
|---|---|---|---|---|---|---|
| T-10047 | 4/2/2026 9:12 (date) | 4/2/2026 11:03 (date) | High | Platform | FALSE | 111 |

Every visible change traces to a specific cleanup step in the Instructions (trim, case normalize, Text to Columns, computed column). No value on the raw sheet has been altered.

## Expected Deliverables

- A single `.xlsx` file with two sheets (`tickets_raw` untouched; `tickets_clean` with the Table).
- A named Excel Table called `Tickets` of exactly 413 rows (420 raw – 7 deduplicated).
- The `OpenedAt` and `ClosedAt` columns typed as dates (not text), confirmed by the `ResolutionMinutes` column producing numeric values with no `#VALUE!` errors.
- A `Severity` column with only High/Medium/Low or blank (no mixed case, no trailing spaces); data validation applied.
- A `SeverityIsMissing` flag column that reads `TRUE` on exactly 19 rows.
- A `ResolutionMinutes` column that returns a positive integer on every row where both timestamps are present.

## Success Criteria

You have completed this lab successfully when you can:

- Point to any ticket ID in the cleaned Table and trace it back to the same row in `tickets_raw` to prove lineage.
- Explain why the missing-severity rows are flagged rather than deleted.
- Explain why Text to Columns was required on the date fields instead of simply changing the cell format.
- Add a new row of fake ticket data at the bottom of the Table and confirm the data validation on Severity and the `SeverityIsMissing` / `ResolutionMinutes` formulas all apply automatically.